(5)

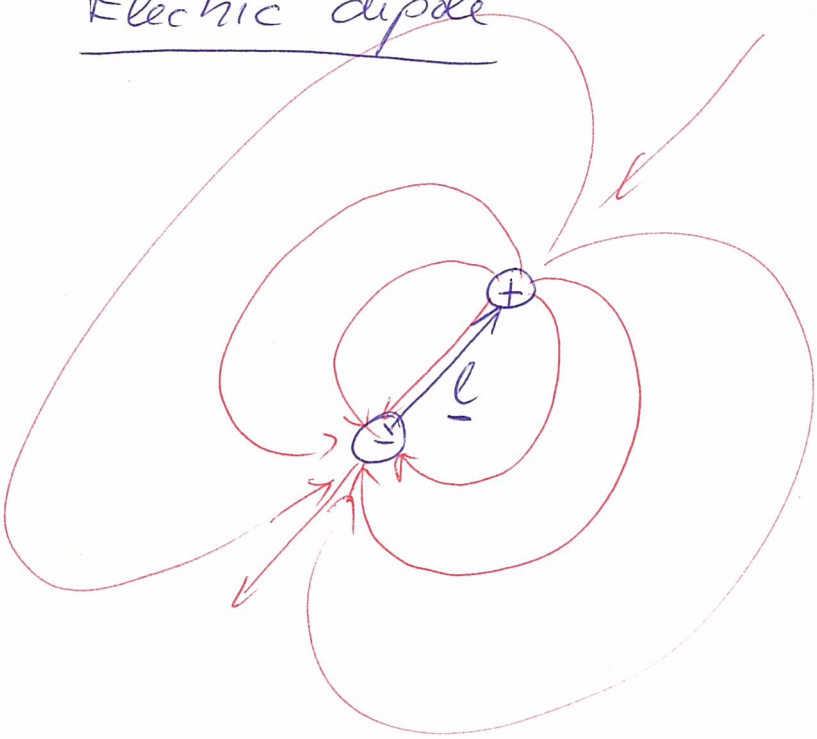
Dielectrics (Insulators)

Experiment:



Insulator in the capacitor decreases its voltage (pot. difference) → increases C

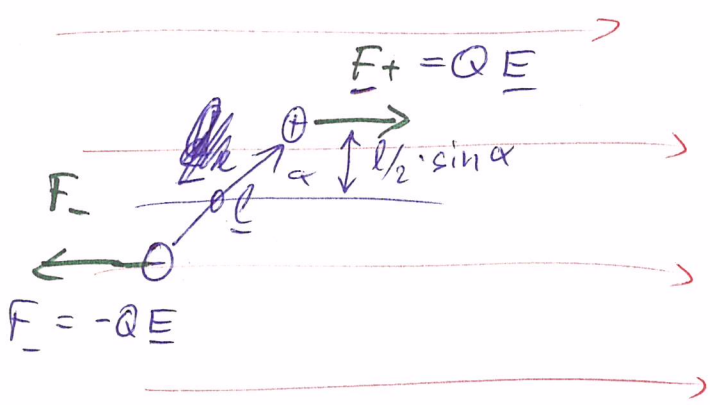
Electric dipole



How to calculate the field of an el. dipole? Superposition pr.!

$$\underline{d_e} := q\,\underline{\ell}$$

Dipole in an homogenous external field



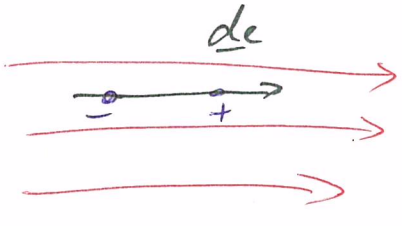
$\sum F = 0$: no acceleration

Torque = force × arm (moment)

$$M = Q\,\underline{E} \cdot \frac{\ell}{2} \sin\alpha \cdot 2$$

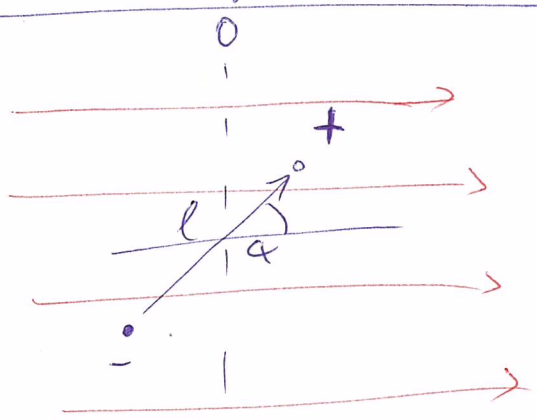
Torque vector

$$\underline{M} = \underline{d_e} \times \underline{E}$$



equilibrium: $\underline{d_e} \parallel \underline{E}$ aligned with the field

Potential energy of the dipole:



superpos princ.

$$U^0_+ = -\int_0^+ \underline{E}\,d\underline{r}\cdot Q = -\frac{E\ell}{2}Q$$

$$U^0_- = -\int_0^- \underline{E}\,d\underline{r}\,(-Q) = -\frac{E(-\ell)}{2}\cdot(-Q) = -\frac{E\ell}{2}Q$$

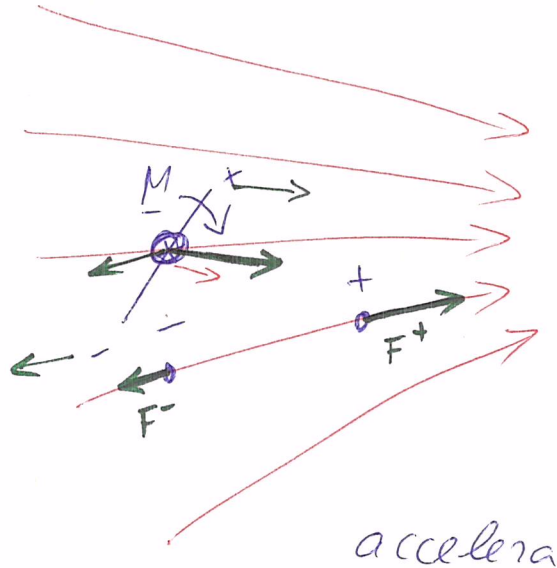
$$U_{pot\pm} = U_- + U_+ = -\underline{E}\cdot\underline{\ell}\cdot Q = -\underline{E}\cdot\underline{d}_e$$

(pot en. negative in equilibrium!

0 when $\perp$

+ when $\underline{d}_e \uparrow\uparrow \underline{E}$ antiparallel.

What happens in an inhomogenous field?



$\underline{F}_d = ?$

$\underline{M}_d = ?$ → aligns with $\underline{E}$

↓

if aligned forces have the same action lines

↓

$F^+ > F^-$

accelerates towards the stronger field

↑

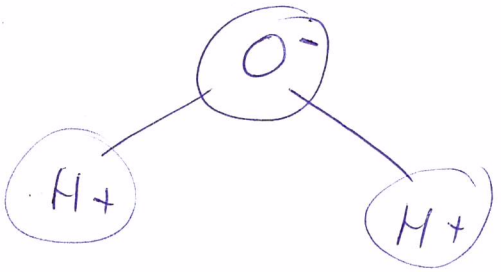
this + peak effect

Experiment?

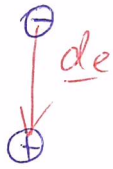
El. wind

Ch 8. Describe the movement of a dipole in inhom field

point like dipoles can be used to model atoms & molecules of insulators!



⇒



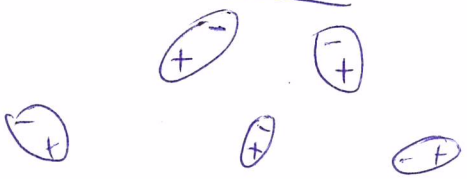
polar molecule have native dipole

Apolar molecules have ∅ dipole if isolated

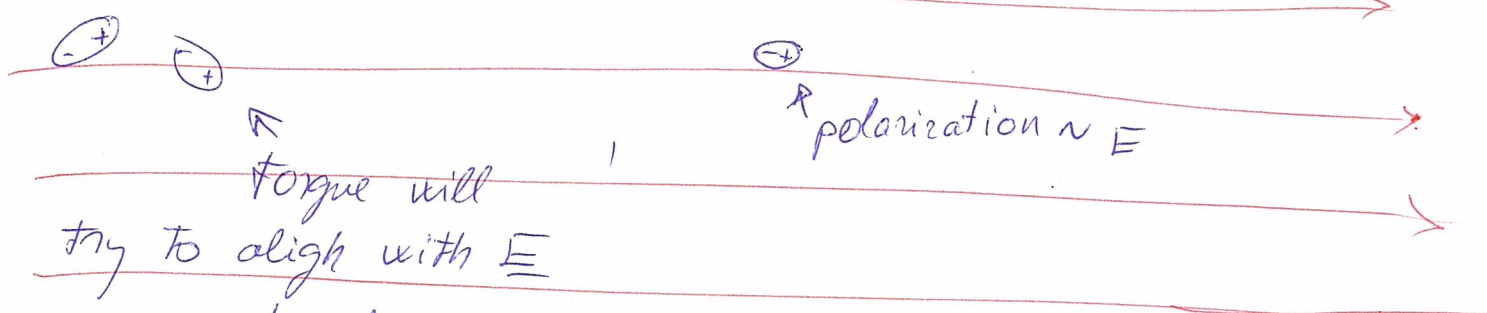
CO_2

$O = C = O$ $\quad d_e = 0$

bulk matter



macroscopic polarization = ∅



polarization ~ E

torque will try to align with $\underline{E}$

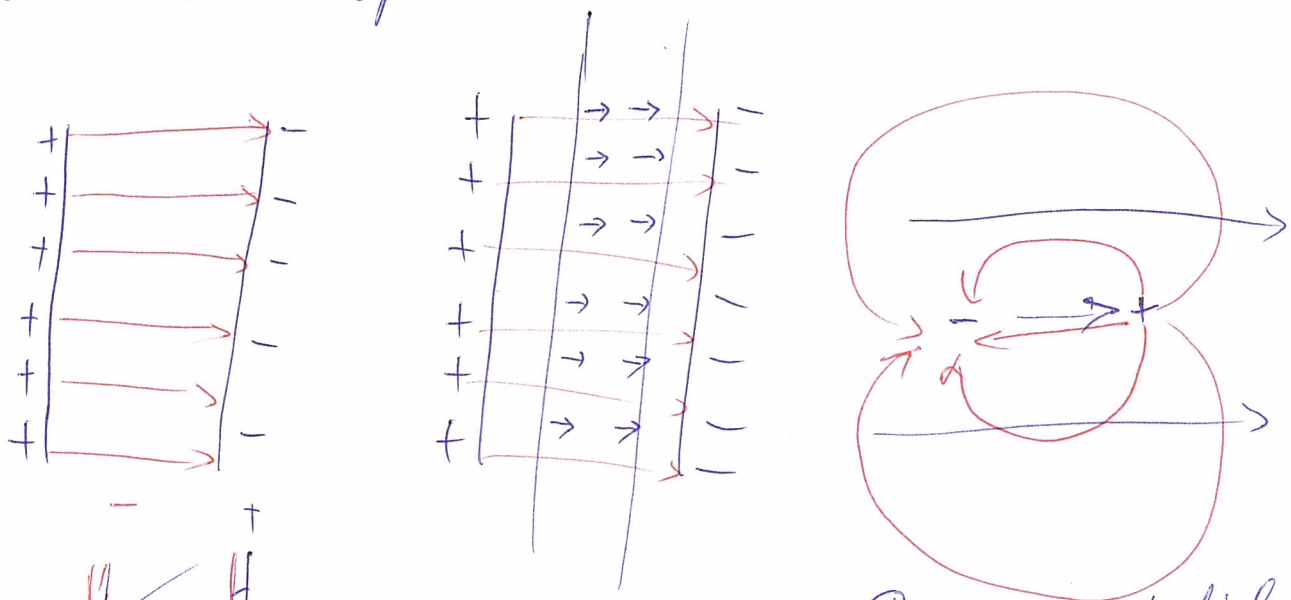
* counter forces:
  - thermal motion
  - dipole dipole interaction!

Our model: we describe the matter with an "average dipole" molecule $\underline{d_e} \sim \underline{E}$

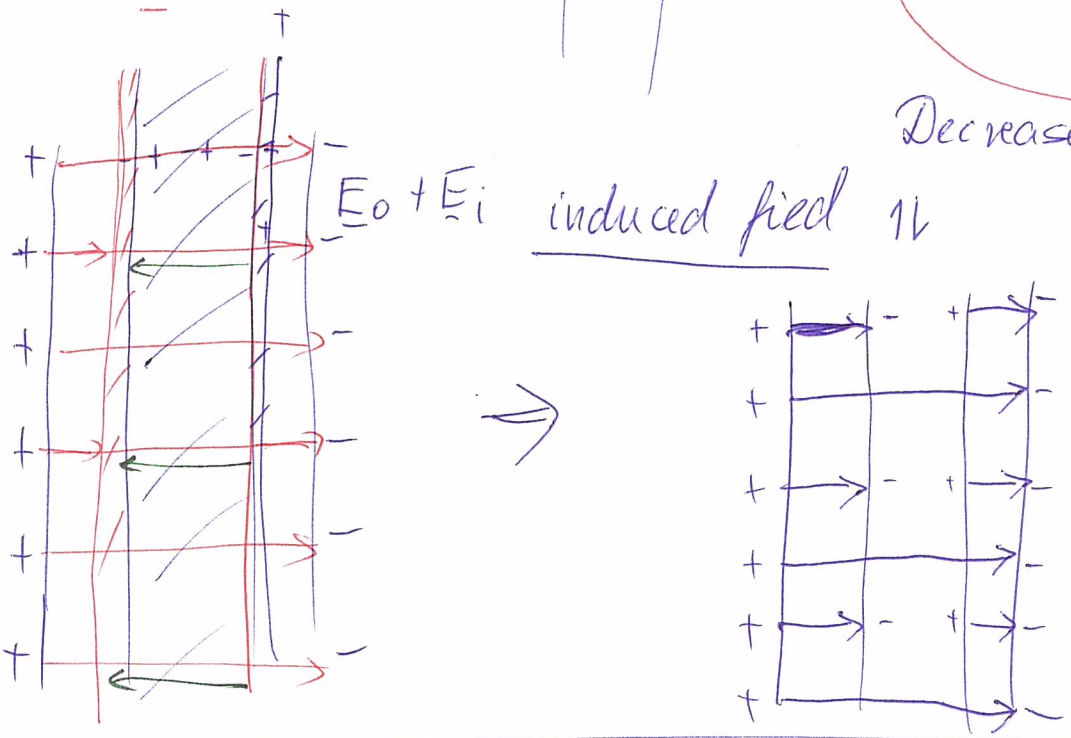
Ch 9
Prove that unordered (random) orientation gives minimal interaction energy in 3D

Note: model is based on averaging of millions of mol. can not be applied for atomic microscopic structures!

We can now understand the effect of the insulator ~~disk~~ in the capacitor:



Decreases ext. field!



$E_0 + E_i$ induced field ↓

What happens to our fundamental laws?

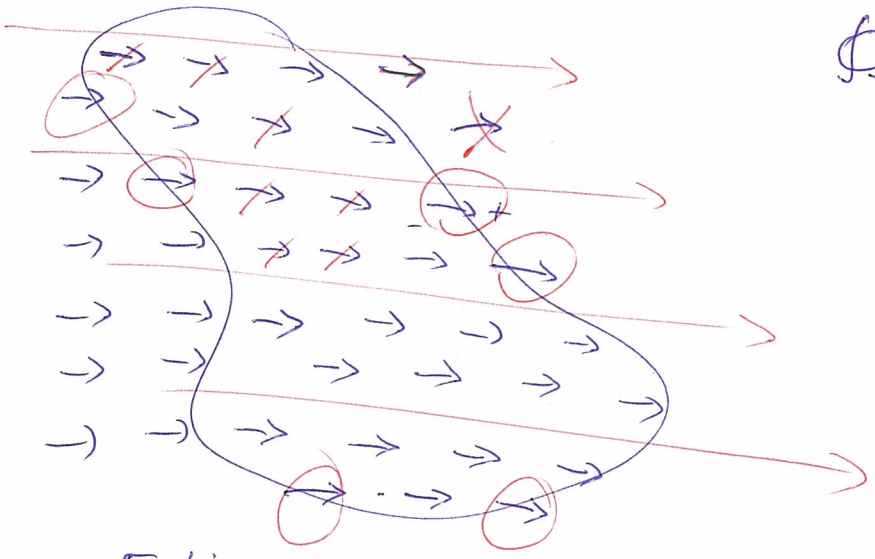
I $\oint \underline{E} d\underline{r} = 0$ ($\operatorname{rot} \underline{E} = 0$) remains the same

II Gauss $\oint \underline{E} d\underline{s} = \frac{Q}{\varepsilon_0} = (Q_f + Q_p)\frac{1}{\varepsilon_0}$

↑ polarizable or bound charges

Let us investigate the effect of bound charges!

Effect of bound charges on Gauss's law



$$\oint \underline{E}\, d\underline{s} = \frac{Q_f}{\varepsilon_0} + \frac{Q_p}{\varepsilon_0}$$ inside

dipoles outside $\oint$

dipoles inside $\oint$

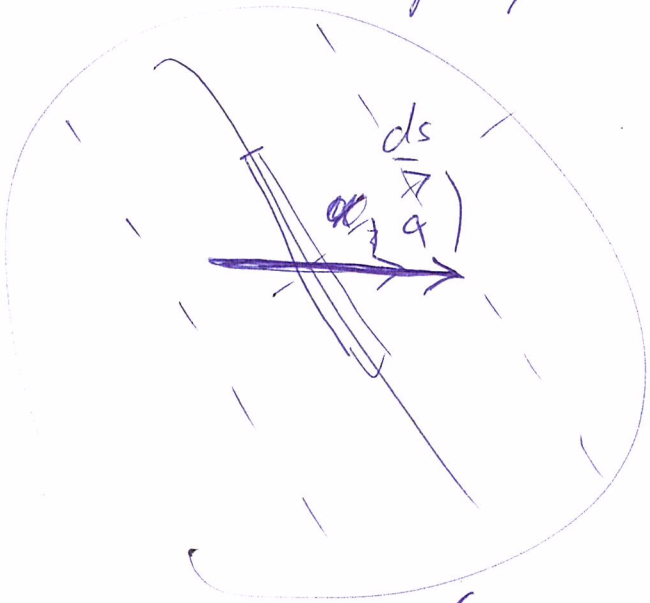
dipoles on the boundary (one end inside the other outside)

Estimate the number of dipoles crossing the boundary surface!

Center of dipole closer to surface than $\frac{l}{2}\cdot\cos\alpha$

~~Thickness~~ of layer $\pm\frac{l}{2}\cos\alpha$

elementary volume $\pm\frac{l}{2}\cos\alpha\cdot ds$



$$dN = n\, l \cos\alpha \cdot ds$$

molecular/atomic (dipole) density $\left[\frac{1}{m^3}\right]$

$$dQ_p = -q\, n\, l \cos\alpha\, ds =$$

($-$: $\cos\alpha > 0$ on right side : $-$ charge ins

$\cos\alpha < 0$ on left $\;\;\;\;$ + charge ins)

$$dQ_p = -\underbrace{n\, q\, l}\cdot d\underline{s} = -\underline{P}\, d\underline{s}$$

$\underline{P}$: dipole moment density or polarization vector

the total p bounded charge: $Q_p = -\oint\!\!\oint \underline{P}\, d\underline{s}$

So Gauss's law:

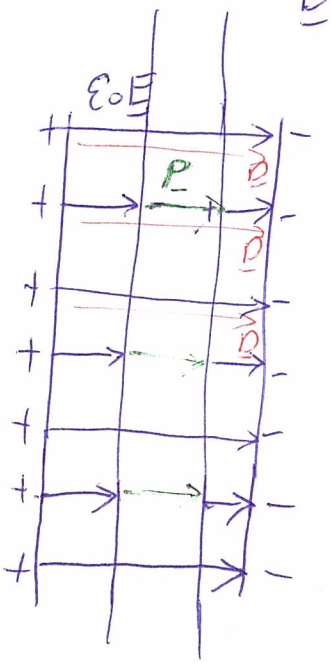
$$\oint_A \underline{E}\,d\underline{s} = \frac{Q_f}{\varepsilon_0} - \frac{1}{\varepsilon_0}\oint \underline{P}\,d\underline{s}$$

$$\oint_A (\varepsilon_0\underline{E} + \underline{P})\,d\underline{s} = Q_f$$

$[D] = \frac{C}{m^2}$

$\underline{D} := \varepsilon_0\underline{E} + \underline{P}$ dielectric displacement vector

$\operatorname{div}\underline{D} = \varrho$



$$\oint (\varepsilon_0\underline{E} + \underline{P})\,d\underline{s} = \boxed{\oint \underline{D}\,d\underline{s} = Q_f} \quad (Q)$$

the D lines (displacement field-lines) start & end on free charges!

the P lines start & end on bound charges (start on − charge!)

($\underline{E}$: all kind of charges)

D is independent of the material in the capacitor, only depends on free charges!

To describe material: we have to know $\underline{P}(\underline{E})$

① Simplest case

Homogenous, isotopic, linear material)

$\underline{P} \sim \underline{E}$ $\quad \underline{P} = \varepsilon_0 \chi \underline{E}$

$\underline{P} \parallel \underline{E}$ $\quad \chi$: dielectric susceptibility

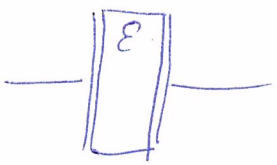
$$\underline{D} = \varepsilon_0\underline{E} + \underline{P} = \varepsilon_0\underline{E} + \varepsilon_0\chi\underline{E} = \varepsilon_0(1+\chi)\underline{E}$$

If the whole observation volume is in homogenous material (isotrop linear), we can simply replace ε_0 by $\varepsilon_0 \varepsilon_r = \varepsilon$

$$\underline{F}_{Coulomb} = \frac{1}{4\pi \varepsilon_0 \varepsilon_r} \frac{Q_1 Q_2}{r^2} \underline{e}_{12}$$

Coulomb force decreasing! $\underline{E}$ is also decreasing!

Capacitor fully loaded with diel:



$$C = \frac{\varepsilon_0 \varepsilon_r}{d} \cdot A$$

Capacitance increasing

$$\underline{D} = \varepsilon_0 \underline{E} + \underline{P} = \varepsilon_0 (1+\chi)\underline{E} = \varepsilon_0 \varepsilon_r \underline{E} = \varepsilon \underline{E}$$

ε_r: relative permittivity ↑ permitt. ~~Fata~~ diel const.

(dep. on freq/f, temp

For ~~most~~ all natural materials $\chi > 0$ normal

$\chi_{air} = 0,00059$ 20 °C

$\chi_{H_2O} = 85 - 87$

$\chi_{glass} = 3,8 - 14$

$\chi_{CaCu_3Ti_4O_{12}}$ 10 000 !!

Specially engineered mat 250 000

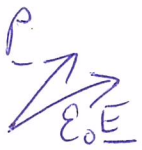
(II) linear anisotropic material

$$\underline{P} = \underline{\underline{\chi}}\, \underline{E}$$

↑ susceptibility tensor (Hermitian-tensor)

funny: in directions other than the eigenvectors,

$\underline{P} \not\parallel \underline{E}$



Leyden cups! (jar?)

(III) Nonlinear behaviour (almost all materials at very high field)

pyro-electric materials: have intrinsic polarization due to crystallin order!

(natural crystals: Tourmaline

GaN, polyvinyl fluorides, $CsNO_3$, $LiTaO_3$

They are also piezoelectric!

Ferroelectric

↑ crystal wit no-centro symmetry

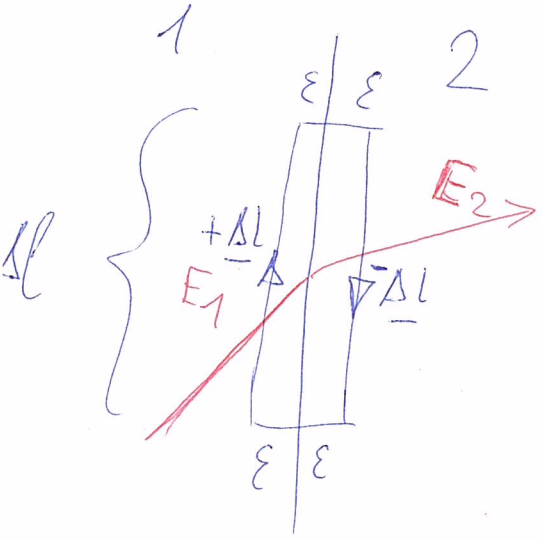
~~CH10~~

sensors

IR Movement ~~detectors~~!

~~Continuity~~ Boundary conditions on the ~~limit~~ of interface 2 materials

I let's apply the first law of static el. on all interface of 2 materials



$$\oint \underline{E}\,\underline{dr} = \phi$$

$$\lim_{\varepsilon \to 0} \oint \underline{E}\,\underline{dr} = \underline{E}_1^{(0)} \cdot \underline{\Delta l} - \underline{E}_2^{(0)} \cdot \Delta l =$$

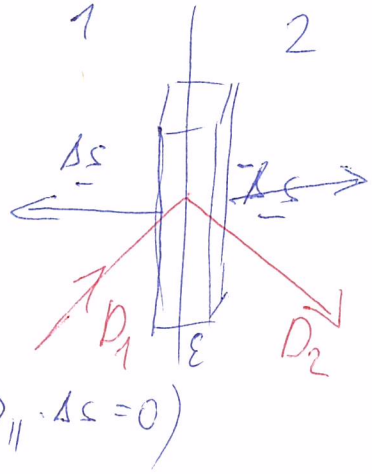
$$= (E_{1T} - E_{2T})\,\Delta l = 0$$

$$E_{1T} = E_{2T}$$

the tangential ~~comp~~ (parallel to surface) component has to be continuous!

The normal component can be different!

II



$(D_{\parallel} \cdot \Delta s = 0)$

Gauss's law

$$\oint \underline{D}\,\underline{ds} = Q_f = \sigma_f \cdot \Delta s$$

$$\underline{D}_1 \cdot \underline{\Delta s} - \underline{D}_2 \underline{\Delta s} = \sigma_f \cdot \Delta s$$

$$D_{1n} - D_{2n} = \sigma_f$$

The discontinuity of the normal comp. of D = surf. ch. dens of field

|| component can be arbitrary

Note:- these conditions are also valid for conductors!
(all materials)

- they remain valid for time-dependent electric fields ⇒ very important for the description of EM wave propagation through the interface of 2 materials